Notes: Yermack (JFE, 1996)

Higher Market Valuation of Companies with a Small Board of Directors

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1 Big picture

- **Question.** Are smaller boards of directors associated with higher firm value and stronger governance outcomes?
- **Main empirical claim.** In a panel of large U.S. industrial corporations from 1984 to 1991, firms with smaller boards have higher Tobin's Q.
- **Central governance hypothesis.** Although larger boards may contain more expertise, they also face coordination, free-riding, slower decision-making, weaker candor, and greater CEO control. The paper tests whether these costs dominate.
- **Primary dependent variable.** The main valuation measure is Tobin's Q, calculated as market value of assets divided by replacement cost of assets.
- **Primary explanatory variable.** Board size is the number of directors serving on the board at the annual meeting date.
- **Main result.** The coefficient on log board size is negative in both OLS and firm fixed-effects specifications. Firms with larger boards have lower market valuation.
- **Nonlinearity.** The board-size valuation relation is convex in the level of board size because the model uses log board size: going from 6 to 12 directors is roughly as costly as going from 12 to 24 directors.
- **Governance mechanism.** Small boards appear to provide stronger CEO incentives: CEO turnover is more sensitive to performance and CEO pay is more sensitive to shareholder wealth when boards are smaller.
- **Supporting evidence.** Smaller boards are associated with better operating ratios, stronger compensation incentives, greater CEO dismissal sensitivity, and positive stock market reactions to large board-size reductions.
- **Takeaway.** Board size is not a neutral governance characteristic. The evidence supports the view that smaller boards are more effective monitors and are valued more highly by investors.

2 Incremental contribution of the paper

2.1 Contribution to corporate governance research

- Earlier governance research focused heavily on board independence, managerial ownership, takeovers, blockholders, and compensation incentives.
- Yermack shifts attention to a simpler and more measurable board design feature: the number of directors.
- The paper provides one of the first large-sample empirical tests of the Lipton–Lorsch and Jensen argument that small boards are more effective.

Interpretation: The paper turns a qualitative governance recommendation–"boards should be small"–into a testable empirical hypothesis using panel data.

2.2 Contribution to board-size debate

- Board size might plausibly help firms by adding expertise, external contacts, and monitoring capacity.
- It might also hurt firms by increasing coordination costs and reducing accountability.
- The paper shows that, in the data, the negative association with valuation dominates.

Interpretation: The paper's contribution is not just documenting board size. It shows that the market values board structure in a way consistent with coordination-cost theories.

2.3 Contribution to mechanism evidence

- The paper does not rely only on Tobin's Q.
- It also studies financial ratios, CEO dismissal sensitivity, CEO pay-performance sensitivity, and stock returns around board-size-change announcements.
- These additional tests connect valuation to plausible channels of monitoring and incentives.

Interpretation: The paper's multi-outcome design is important because it helps distinguish the governance interpretation from a purely mechanical correlation between board size and firm type.

3 Data and measurement

3.1 Sample construction

- The sample contains 452 large U.S. public corporations.
- The sample period is 1984–1991.
- Firms are included if they are ranked by *Forbes* as among the 500 largest U.S. public corporations at least four times during the eight-year period.
- Utility and financial firms are excluded.
- The final panel contains 3,438 firm-year observations in the main board-characteristics table.

Interpretation: The sample focuses on large industrial corporations where boards are observable, economically important, and comparable across firms.

3.2 Board variables

- **Board size:** number of directors sitting on the board as of the annual meeting date.
- **Inside directors:** current or former officers of the firm.
- **Gray directors:** directors with personal or business ties to the firm, but not current or former officers.
- **Outside directors:** directors who are neither insiders nor gray directors.
- **CEO-directors:** outside directors who are CEOs of other firms.

Interpretation: The paper separates board size from board composition. This is important because a large board could simply reflect more outside directors, insiders, gray directors, or CEO-directors.

3.3 Governance and ownership variables

- CEO from founding family dummy.
- Non-CEO chairman of board dummy.
- Presence of a 5% stockholder-director dummy.
- Director and officer ownership as a percentage of common equity.
- Director fees and director turnover.

Interpretation: These variables allow the paper to test whether board size is only proxying for other governance arrangements.

3.4 Firm value and performance measures

- The main valuation measure is Tobin's Q.
- Operating performance controls include return on assets and lagged returns on assets.
- Other firm-level controls include firm size, capital expenditures over sales, diversification measured by business segments, and year and industry fixed effects.
- Additional outcomes include sales/assets, return on assets, return on sales, CEO turnover, CEO compensation changes, and event-study abnormal returns.

Interpretation: The paper uses Tobin's Q for valuation, but then checks whether the implied governance interpretation is visible in accounting performance and CEO incentive mechanisms.

4 Empirical specifications

4.1 Baseline valuation regression

The main regression is:

$$Q_{it} = \alpha + \beta \log(BoardSize_{it}) + \Gamma' X_{it} + \varepsilon_{it}. \tag{1}$$

The richer panel version is:

$$Q_{it} = \alpha_i + \lambda_t + \beta \log(BoardSize_{it}) + \Gamma' X_{it} + \varepsilon_{it}, \tag{2}$$

where α_i are firm fixed effects and λ_t are year fixed effects.

Interpretation: The OLS model compares firms to other firms. The fixed-effects model asks whether the same firm has lower valuation in years when its board is larger.

4.2 Controls in the valuation model

The control vector includes:

- current and lagged return on assets;
- firm size measured by log total capital;
- capital expenditures over sales;
- number of business segments;
- percentage of outside directors;
- officer and director stock ownership;
- year indicators and industry controls.

Interpretation: The controls address the most obvious alternative explanations: small boards could be correlated with small firms, high-growth firms, focused firms, higher inside ownership, or different industry structures.

4.3 Board-change causality test

To test whether past performance drives board-size changes, the paper estimates models of director appointments, departures, and net board-size changes:

$$BoardChange_{it} = \alpha + \beta_1 AbnormalReturn_{it} + \beta_2 AbnormalReturn_{i,t-1} + \Gamma' Z_{it} + \varepsilon_{it}. \tag{3}$$

Interpretation: If poor performance mechanically caused firms to add directors, the valuation result could be reverse causality. Table 3 finds little support for that interpretation.

4.4 CEO dismissal incentives

The CEO dismissal model is a probit:

$$Pr(CEOExit_{it} = 1) = \Phi\left(\alpha + \beta Performance_{it} + \delta \log(BoardSize_{it}) \times Performance_{it} + \Gamma' X_{it}\right). \tag{4}$$

Interpretation: A positive interaction between poor performance and small board effectiveness means that smaller boards are more willing to dismiss CEOs after poor performance.

4.5 CEO compensation incentives

CEO pay-performance sensitivity is estimated with:

$$\Delta Pay_{it} = \alpha + \beta \Delta ShareholderWealth_{it} + \delta \Delta ShareholderWealth_{it} \times \log(BoardSize_{it}) + \Gamma' X_{it} + \varepsilon_{it}. \tag{5}$$

Interpretation: A negative interaction means CEO compensation is less performance-sensitive when boards are larger.

4.6 Board-size announcement returns

The event study examines companies announcing large board-size changes. Abnormal returns are measured using:

- raw return minus market return;
- market-model abnormal return;
- CRSP excess return.

Interpretation: Positive returns around board-size reductions provide direct market evidence that investors value smaller boards.

5 Identification and interpretation

5.1 Main identification challenge

Board size is not randomly assigned. Firms choose board structure based on size, complexity, age, ownership, and past performance. The central concern is therefore omitted variable bias or reverse causality.

Interpretation: The paper's results should be interpreted as robust evidence consistent with small-board effectiveness, not as a randomized experiment.

5.2 How the paper addresses confounding

- It controls for firm size, profitability, growth opportunities, business segments, board composition, and ownership.
- It uses firm fixed effects to exploit within-firm variation.
- It estimates board-change models to test whether past performance predicts changes in board size.
- It studies mechanisms: financial ratios, CEO dismissal incentives, pay-performance sensitivity, and announcement returns.

Interpretation: The strongest evidence comes from the consistency across valuation, operating performance, and incentive outcomes.

5.3 Limits of the design

- Board size remains endogenous.
- The sample contains large U.S. corporations, so external validity to small firms or financial firms is limited.
- The event-study sample is small, with only ten large board-size-change announcements.
- Tobin's Q may reflect unobserved investment opportunities as well as governance quality.

Interpretation: The paper is influential because it established a robust empirical regularity, even though later research must still address causal identification more directly.

6 Tables and figures

6.1 Table 1 and Figure 1: board characteristics and valuation pattern

Read:

- Table 1 shows that the mean board has 12.25 directors and the median board has 12 directors.
- Inside directors make up 36% of the board on average, gray directors 10%, and outside directors 54%.
- Board size is positively correlated with director fees and turnover but negatively correlated with CEO-founder status, 5% stockholder-director presence, and officer/director ownership.
- Figure 1 shows that both mean and median Tobin's Q decline as board size rises.
- The decline is steepest when moving from small boards to medium-sized boards.

Economic message: The descriptive evidence already points to the paper's main result: firms with smaller boards are valued more highly by the market.

Table 1
Board of directors characteristics

Descriptive statistics for characteristics of boards of directors. The sample consists of 3,438 annual observations for 452 companies between 1984 and 1991. Companies are included in the sample if they are ranked by *Forbes* magazine as one of the 500 largest U.S. public corporations at least four times during the eight-year sample period. Utility and financial companies are excluded. The table presents the mean, median, and standard deviation for each variable, as well as Pearson sample correlation coefficients between the board-size variable and all others.

Board size represents the number of members of the board of directors as of the annual meeting date during each fiscal year. The percentage of inside directors is the fraction of board members who are current or former officers of each company. Gray directors are those who have substantial business relationships with the company, either personally or through their main employers, and also relatives of corporate officers. Outside directors are those who have neither inside nor gray status. Director fees include annual retainers and fees paid for regular and special board meetings during the fiscal year. The dummy variable for director stock option plan equals one if the company has a plan in place for awarding stock options to outside directors. Director turnover is the fraction of board members who leave before the next annual meeting. The CEO-from-founding-family dummy variable equals 1 if the CEO is from a family which either founded the company or acquired control during a takeover. The dummy variable for 5% stockholder-directors equals 1 if one or more members of the board beneficially own at least 5% of the common stock or serve as representatives of an outside 5% holder (not including employee stock ownership plans).

	Mean	Median	Std. dev.	Correlation with board size
Board size	12.25	12	0.43	1.00
Board composition				
Inside directors	0.36	0.33	0.16	− 0.09
Gray directors	0.10	0.08	0.12	− 0.16
Outside directors	0.54	0.56	0.19	0.17
Directors who are CEOs of other firms	0.14	0.13	0.12	0.14
Compensation and turnover				
Director fees (1991 dollars)	$29,539	$29,601	$10,657	0.33
Director stock option plan (dummy variable)	0.09	0	0.29	− 0.09
Director turnover (% of board per year)	8.3%	7.1%	10.2%	0.15
Governance structure and stock ownership				
CEO from founding family (dummy variable)	0.24	0	0.42	− 0.26
Non-CEO chairman of board (dummy variable)	0.17	0	0.38	− 0.06
Presence of 5% stockholder-director (dummy variable)	0.24	0	0.43	− 0.14
Director and officer stock ownership (% of common)	9.1%	2.8%	14.3%	− 0.27

Last column: All correlations with board size are significant at the 1% level.

Table 1: board characteristics

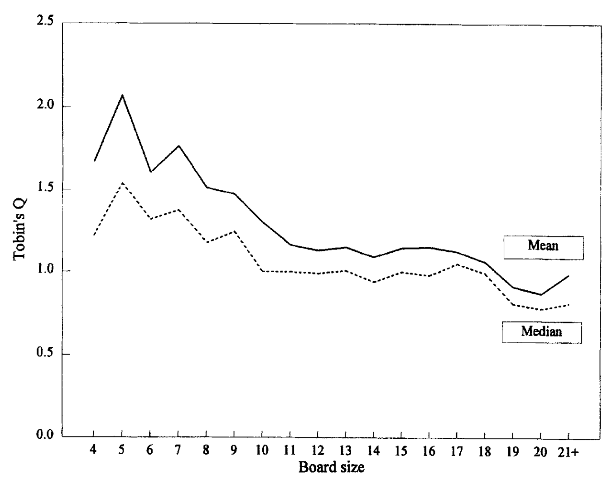

Fig. 1. Board size and Tobin's Q: Sample means and medians.

Figure 1: board size and Tobin's Q

6.2 Tables 2 and 3: valuation regression and reverse-causality test

Read:

- Table 2 estimates the association between board size and Tobin's Q.
- The OLS coefficient on log board size is −0.428 and the fixed-effects coefficient is −0.337.
- Both are statistically significant at the 1% level.

- Table 3 tests whether past performance predicts director appointments, director departures, and board-size changes.
- Poor performance predicts more director turnover, but the evidence does not show that poor performance causes firms to expand board size.

Economic message: The valuation relation is robust, and the reverse-causality test does not support the view that low-value firms simply add directors after bad performance.

Table 2
Regression coefficient estimates: Board size and market valuation

Regression coefficient estimates of the association between Tobin's Q and the number of directors sitting on company boards. The sample consists of 3,438 annual observations for 452 firms between 1984 and 1991. Companies are included in the sample if they are ranked by *Forbes* magazine as one of the 500 largest U.S. public corporations at least four times during the eight-year sample period. Utility and financial companies are excluded. The dependent variable is an estimate of Tobin's Q at the end of each fiscal year. The log of board size is the natural log of the number of directors sitting on each company's board as of the annual meeting date each year, as reported in company proxy statements. The first column presents OLS estimates with White (1980) robust standard errors. The second column presents estimates from a fixed-effects model, which assigns a unique intercept to each company.

The model includes control variables for company performance (return on assets in the current year and two lags), firm size (the log of total capital, in 1991 dollars), growth opportunities (capital expenditures over sales), diversification (the number of business segments for which financial statement data is reported), board composition (the percentage of outside directors on the board), and inside stock ownership (director and officer beneficial ownership, in percent). *ROA* equals operating income over total assets (start of year) and is compounded continuously. Total capital equals the market value of common stock at the end of the year, plus estimates of the market values of long-term debt and preferred stock. Both models include dummy variables for years, and the OLS model includes two-digit SIC industry dummy variables.

Dependent variable: *Tobin's Q* Variable	OLS estimates	Fixed-effects estimates
Log of board size	− 0.428***	− 0.337***
	(0.043)	(0.056)
Return on assets (current year)	3.856***	2.048***
	(0.403)	(0.147)
Return on assets (prior year)	0.502	− 0.093
	(0.536)	(0.151)
Return on assets (two years prior)	1.039**	0.450***
	(0.451)	(0.130)
Firm size (log of total capital)	0.119***	0.413***
	(0.012)	(0.020)
Capital expenditures/Sales	− 0.116	0.176
	(0.209)	(0.123)
Number of business segments	− 0.042***	− 0.049***
	(0.006)	(0.009)
Board composition (% outside directors)	− 0.213***	0.172*
	(0.067)	(0.088)
Officer and director stock ownership (%)	0.279***	0.310***
	(0.096)	(0.108)
Sample size	3,400	3,400
F-statistic	68.1	2.7
(P-value)	(0.00)	(0.00)
R-squared	0.5459	0.3021

Significant at 1% (***), 5% (**), and 10% (*) levels.

Table 2: board size and market valuation

Table 3
Regression coefficient estimates: Effect of company performance on director appointments, departures, and changes in board size

Coefficient estimates for regression models of changes in board size. The first two columns present maximum-likelihood Poisson models of the number of directors joining and leaving each company's board. The third column presents OLS estimates of the net change in board size, equal to director additions minus director departures. All variables are measured annually, based upon the membership of companies' boards as reported in proxy statements for annual shareholder meetings.

The sample consists of 3,438 annual observations for 452 firms between 1984 and 1991. Companies are included in the sample if they are ranked by *Forbes* magazine as one of the 500 largest U.S. public corporations at least four times during the eight-year sample period. Utility and financial companies are excluded. The key explanatory variable for all three-models is the firm's abnormal stock return during the fiscal year, defined as the raw return minus the return predicted by the CAPM. Other explanatory variables are similar to those used by Hermalin and Weisbach (1988). CEO at retirement age is a dummy variable equal to one if the company's CEO is between the ages of 62 and 66. New CEO is a dummy equal to one if the CEO has four years of tenure or less. All models include two-digit SIC industry dummy variables. The text discusses assumptions used in calculating abnormal stock returns. Each coefficient estimate appears with robust standard errors.

Dependent variable	Director appointments	Director departures	Change in board size
Estimation	Poisson ML	Poisson ML	OLS
	Estimate	Estimate	Estimate
Abnormal stock return	− 0.238**	− 0.238***	− 0.005
(fiscal year)	(0.093)	(0.084)	(0.106)
Abnormal stock return	− 0.087	− 0.126	0.039
(prior fiscal year)	(0.088)	(0.084)	(0.091)
Change in firm size	0.419***	− 0.347*	0.770***
[log(sales)]	(0.163)	(0.185)	(0.198)
CEO at retirement age	0.129**	0.133**	− 0.015
(dummy for ages 62 to 66)	(0.054)	(0.055)	(0.062)
New CEO	0.148***	0.273***	− 0.156***
(dummy for 4 years' tenure or less)	(0.048)	(0.050)	(0.055)
SIC industry dummies	2-digit	2-digit	2-digit
Sample size	2,943	2,943	2,943
F-statistic			1.9
(P-value)			(0.00)
R-squared			0.0313

Significant at 1% (***), 5% (**), and 10% (*) levels.

Table 3: performance and board changes

6.3 Tables 4 and 5: operating ratios and CEO dismissal incentives

Read:

- Table 4 shows that larger boards are associated with worse financial ratios.
- The coefficient on log board size is negative for sales/assets, return on assets, and return on sales.
- Table 5 shows that CEO turnover is more sensitive to poor performance in firms with smaller boards.
- The interaction between abnormal stock return and log board size is positive, meaning that performance-related dismissal sensitivity declines as boards become larger.

Economic message: Smaller boards appear not only to be valued more highly, but also to monitor and discipline management more effectively.

Table 4
Fixed-effects estimates: Board size and financial ratios

Coefficient estimates for fixed-effects regression models of financial ratios. Standard errors appears below each estimate. The sample consists of 3,438 annual observations for 452 firms between 1984 and 1991. Companies are included in the sample if they are ranked by *Forbes* magazine as one of the 500 largest U.S. public corporations at least four times during the eight-year sample period. Utility and financial companies are excluded.

The dependent variables are three standard measures of operating efficiency and profitability. *ROS* and *ROA* are based on operating income and are compounded continuously. The main explanatory variable for each model is the natural log of board size, which represents the number of directors sitting on each company's board as of the annual meeting for each fiscal year. In addition to the other controls listed, each model includes dummy variables for individual years.

Dependent variable	Sale/ assets	Return on assets	Return on sales
	Estimate	Estimate	Estimate
Log of board size	− 0.163**	− 0.037***	− 0.025***
	(0.079)	(0.008)	(0.006)
Board composition (% outside directors)	− 0.187	0.005	− 0.009
	(0.126)	(0.012)	(0.009)
Officer and director stock ownership (%)	0.287*	0.022	− 0.029***
	(0.153)	(0.015)	(0.011)
Number of business segments	0.006	− 0.0026**	− 0.0020**
	(0.013)	(0.0013)	(0.0009)
Firm size (log of total capital)	0.083***	0.045***	0.031***
	(0.026)	(0.003)	(0.002)
Sample size	3,428	3,425	3,425
F-statistic	23.6	66.3	44.3
(*P*-value)	(0.00)	(0.00)	(0.00)

Table 4: board size and financial ratios

Table 5
Probit coefficient estimates: Board size and CEO dismissal incentives

Regression coefficient estimates for a binary probit model of CEO turnover. The dependent variable equals one if a CEO leaves his position during the second half of the current fiscal year or the first half of the subsequent fiscal year. The main explanatory variable is the firm's cumulative abnormal stock return for the current year and two prior years. The abnormal stock return is defined as the raw stock return minus the return predicted by the CAPM. The sample consists of 3,438 annual observations for 452 firms between 1984 and 1991. Companies are included in the sample if they are ranked by *Forbes* magazine as one of the 500 largest U.S. public corporations at least four times during the eight-year sample period. Utility and financial companies are excluded. Coefficient estimates appear with White (1982) robust standard errors. Missing values occur for companies that do not have a three-year history of common stock returns.

To illustrate how CEO incentives from the threat of dismissal are affected by different characteristics of boards of directors, the second column presents estimates for a model that includes an interaction term between the abnormal stock return and the log of board size. Both models also include variables for the fraction of common stock owned by the CEO, CEO age, dummy variables for CEO ages 64, 65, and 66, and dummies for one-digit SIC industries.

Dependent variable: *CEO leaves position (0, 1)*	Estimate	Estimate
CEO age	0.043***	0.043***
	(0.006)	(0.006)
CEO stock ownership (%)	− 2.022***	− 2.027***
	(0.755)	(0.746)
Cumulative abnormal stock return (current year and two prior years)	− 0.358***	− 1.496***
	(0.071)	(0.581)
Interaction term: Abnormal stock return × log (board size)		0.471**
		(0.239)
Sample size	3,305	3,305

Significant at 1% (***), 5% (**), and 10% (*) levels.

Table 5: board size and CEO dismissal incentives

6.4 Table 6: CEO compensation incentives

Read:

- Table 6 estimates CEO pay-performance sensitivity using the annual change in salary and bonus.
- The basic pay-performance coefficient is positive.
- The interaction between shareholder wealth changes and log board size is negative and statistically significant.
- This implies that CEO compensation is less performance-sensitive when boards are larger.

Economic message: Smaller boards provide stronger compensation incentives. This reinforces the monitoring mechanism behind the valuation results.

Table 6
OLS estimates: Board size and CEO compensation incentives

OLS estimates for a model of CEO performance incentives from compensation. The dependent variable is the annual change in the CEO's salary and bonus. The main explanatory variable is the annual change in stockholder wealth. To illustrate how performance incentives are affected as board size increases, the model in the right column includes an interaction term between the change in stockholder wealth and the log of board size.

The sample consists of 3,438 annual observations for 452 firms between 1984 and 1991. Companies are included in the sample if they are ranked by *Forbes* magazine as one of the 500 largest U.S. public corporations at least four times during the eight-year sample period. Utility and financial companies are excluded. Coefficient estimates appear with White (1980) robust standard errors. A large number of missing values occur due to the first-differencing framework (which eliminates one year of data) and episodes of CEO turnover (each of which eliminate two first-difference observations).

Dependent variable: *Change in CEO's salary + bonus*	Estimate	Estimate
Constant	39,348***	35,668***
	(11,842)	(12,750)
Change in stockholder wealth (per $1,000)	0.015**	0.139**
	(0.007)	(0.065)
Interaction term:		− 0.045**
Change in stockholder wealth × log (board size)		(0.023)
Sample size	2,412	2,412
F-statistic	8.0	6.6
(P-value)	(0.00)	(0.00)
R-squared	0.0033	0.0055

Significant at 1% (***), 5% (**), and 10% (*) levels.

6.5 Table 7: market reactions to large board-size changes

Read:

- Table 7 studies ten large board-size-change announcements.
- Companies decreasing board size earn positive abnormal returns around the announcement.
- Companies increasing board size earn negative abnormal returns.
- The difference in means is positive and statistically significant across alternative abnormal-return calculations.

Economic message: The event-study evidence provides direct support for the market valuation interpretation: investors react favorably to large board reductions and unfavorably to large board expansions.

Table 7
Reactions to announcements of significant board size changes

Investor reactions to announcements by ten sample companies of significant changes in board size. Companies were identified by searches of the Nexis database and the Wall Street Journal Index. The analysis includes firms that announced changes in board size of at least four members on an event date in 1984 or later. Events are excluded if board size changes were related to mergers, CEO transitions, or corporate control contests.

Abnormal stock returns around the announcement dates are calculated by three methods: raw returns minus market returns, raw returns minus market model expected returns, and excess returns extracted from the CRSP database. The market return used in calculations is the CRSP value-weighted index, and market model parameters are estimated from one year of daily trading data prior to the event period. T-statistics for market model abnormal returns are based on standardized prediction errors (Dodd and Warner, 1983).

	Number of directors		% outside directors		Abnormal stock returns 1 day before to 1 day after announcement		
Companies decreasing board size	Before	After	Before	After	Net-of-market	Market model	CRSP excess return
Chrysler (1991)	20	13	70%	77%	5.1%	5.4%	5.5%
Control Data (1985)	18	14	61%	36%	5.9%	7.2%	6.4%
Adolph Coors (1988)	10	5	0%	0%	0.6%	0.7%	n.a.
W.R. Grace (1991)	33	26	36%	54%	− 0.7%	− 1.8%	0.3%
Perkin–Elmer (1991)	13	9	85%	89%	2.1%	2.1%	1.6%
Time Warner (1992)	21	15	48%	67%	0.7%	0.2%	0.5%
Mean					2.3%*	2.3%	2.9%*
T-statistic					2.10	1.67	2.24
P-value					0.09	0.17	0.09

Table 7, part 1: board-size decreases

	Number of directors		% outside directors				
Companies increasing board size	Before	After	Before	After	Net-of-market	Market model	CRSP excess return
Cummins Engine (1985)	14	18	50%	50%	− 1.1%	− 0.7%	− 1.2%
Lafarge (1985)	10	16	30%	50%	− 7.9%	− 7.5%	− 7.0%
Squibb (1984)	17	22	35%	45%	− 0.5%	− 0.3%	− 1.2%
Xerox (1987)	14	18	50%	50%	− 1.5%	− 1.3%	− 1.7%
Mean					− 2.8%	− 2.5%	− 2.8%
T-statistic					− 1.59	− 1.44	− 1.94
P-value					0.21	0.25	0.15
Difference in means					5.0%**	4.8%*	5.6%**
T-statistic					2.46	2.17	2.95
P-value					0.04	0.06	0.02
					Abnormal stock returns 10 days before to 10 days after announcement		
Difference in means					17.6%***	17.7%***	14.9%**
T-statistic					3.77	3.41	3.31
P-value					0.01	0.01	0.01

Significant at 1% (***), 5% (**), and 10% (*) levels.

Table 7, part 2: board-size increases

7 Detailed result synthesis

7.1 Valuation result

The paper's central empirical result is that board size is negatively related to Tobin's Q. The result is present in both cross-sectional and within-firm specifications. The log specification implies that the valuation cost is larger when moving from a very small board to a medium board than when adding one more director to an already large board.

7.2 Mechanism through monitoring

The governance mechanism is supported by CEO turnover and compensation results. Smaller boards dismiss CEOs more readily after poor performance and tie pay more closely to shareholder wealth. These two channels suggest that smaller boards have stronger monitoring capacity.

7.3 Operating performance

The negative relation between board size and financial ratios indicates that the valuation result is connected to real operating outcomes. Larger boards are associated with lower efficiency and profitability.

7.4 Market reaction evidence

The board-size-change announcement study is small but directionally important. Investors react positively to board reductions and negatively to board expansions, consistent with the view that board size has causal value implications.

8 Short summary

- The paper studies 452 large U.S. industrial corporations from 1984 to 1991.
- The average board has about 12 directors.
- Tobin's Q declines as board size increases.
- The result survives OLS, firm fixed effects, and controls for size, profitability, growth, diversification, board composition, and ownership.
- Past performance does not explain board-size changes in a way that overturns the result.
- Larger boards are associated with worse financial ratios.
- Smaller boards make CEO turnover more performance-sensitive.
- Smaller boards make CEO compensation more performance-sensitive.
- Investors react positively to large board-size decreases and negatively to large board-size increases.
- The paper provides early evidence that smaller boards are more effective governance structures.

9 Conclusion

9.1 Core conclusion

Yermack finds a robust inverse relation between board size and firm value. The evidence supports the view that smaller boards are more effective monitors and that large boards impose coordination costs that reduce firm value.

9.2 Economic intuition

A board with too many members can suffer from free-riding, slow deliberation, weak accountability, and reduced candor. CEOs may also find large boards easier to control. Smaller boards are better able to monitor, replace, and incentivize top executives.

9.3 Takeaway

The paper's central lesson is that board structure matters. The number of directors is not a cosmetic governance feature; it is associated with valuation, operating performance, CEO incentives, and market reactions to governance changes.